

THERMAL ENGINEERING LAB

HEAT ENGINE LAB

CYCLE – 1

1. PERFORMANCE TEST ON FOUR STROKE SINGLE CYLINDER DIESEL ENGINE
2. PERFORMANCE TEST ON FOUR STROKE FOUR CYLINDER DIESEL ENGINE
3. PERFORMANCE TEST ON FOUR STROKE THREE CYLINDER PETROL ENGINE
4. PERFORMANCE TEST ON AIR BLOWER TEST RIG
5. PERFORMANCE TEST ON SINGLE STAGE RECIPROCATING AIR COMPRESSOR TEST RIG

LAB IN CHARGE

Professor : Dr. Gireesh Kumaran Thampi

1. PERFORMANCE TEST ON SINGLE CYLINDER 4 STROKE DIESEL ENGINE

Introduction

A machine, which uses heat energy obtained from combustion of fuel and converts it into mechanical energy, is known as a Heat Engine. They are classified as External and Internal Combustion Engine. In an External Combustion Engine, combustion takes place outside the cylinder and the heat generated from the combustion of the fuel is transferred to the working fluid which is then expanded to develop the power. An Internal Combustion Engine is one where combustion of the fuel takes place inside the cylinder and converts heat energy into mechanical energy. IC engines may be classified based on the working cycle, thermodynamic cycle, speed, fuel, cooling, method of ignition, mounting of engine cylinder and application.

Diesel Engine is an internal combustion engine, which uses heavy oil or diesel oil as a fuel and operates on two or four stroke. In a 4-stroke Diesel engine, the working cycle takes place in two revolutions of the crankshaft or 4 strokes of the piston. In this engine, pure air is sucked to the engine and the fuel is injected with the combustion taking place at the end of the compression stroke. The power developed and the performance of the engine depends on the condition of operation. So it is necessary to test an engine for different conditions based on the requirement.

Specification

Engine	: Four stroke single cylinder
Make	: Kirloskar
Power	: 5 HP
Engine speed	: 1500
Fuel	: Diesel
Bore	: 80 mm
Stroke length	: 110 mm

Observations and calculations:

[illegible]

Starting : Cranking

Working cycle : Four stroke

Method of cooling : Water cooled

Method of ignition : Compression ignition

Description of the apparatus:

The test rig is built for loading mentioned below:

1. The equipment consists of KIRLOSKAR Diesel Engine (Crank started) of 5hp (3.7kW) capacity and is Water cooled. The Engine is coupled to a Rope Brake Drum Dynamometer for loading purposes. Coupling is done by an extension shaft in a separate bearing house. The dynamometer is connected to the spring load assembly for varying the load.
2. Thermocouples are provided at appropriate positions and are read by a digital temperature indicator with channel selector to select the position.
3. Rota meters are used for direct measurement of water flow rate to the engine and calorimeter respectively.
4. Engine Speed and the load applied at various conditions is determined by a Digital RPM Indicator and spring balance reading.
5. A separate air box with orifice assembly is provided for regularizing and measuring the flow rate of air. The pressure difference at the orifice is measured by means of Manometer.
6. A volumetric flask with a fuel distributor is provided for measurement and directing the fuel to the engine respectively.

Equations used:

Brake power

$$BP = \frac{2 \pi N (W - S) \left(\frac{D+d}{2} \right) 9.81}{60000} \quad (in \text{ kW})$$

Where,

S is the spring balance reading in kg

W is the dead weights in kg

D is the diameter of the brake drum = 0.33m

Calculations:

Brake power

$$BP = \frac{2 \pi N (W - S) \left(\frac{D+d}{2} \right) 9.81}{60000} = \text{_____} kW$$

Mass of fuel consumed

$$mfc = \frac{0.82 \times 3600 \times X}{1000 t} = \text{_____} kg/hr$$

Specific fuel consumption

$$sfc = \frac{mfc}{BP} = \text{_____} kg/kW hr$$

Actual volume of air sucked into the cylinder

$$V_a = C_d A \sqrt{2 g H} 3600 = \text{_____} m^3/hr$$

$$H = \frac{h \rho_w}{1000 \rho_a} = \text{_____}$$

Swept volume

$$V_s = \left(\frac{\pi d^2}{4} L \frac{N}{2} 60 \right) = \text{_____}$$

Volumetric efficiency

$$\eta_v = \frac{V_A}{V_s} 100 = \text{_____} \%$$

Brake thermal efficiency

$$\eta_{bth} = \frac{BP 3600 \times 100}{mfc \times CV} = \text{_____}$$

$$CV = 45200 \text{ kJ/kg}$$

Mechanical efficiency

$$\eta_{mech} = \frac{BP}{IP} = \text{_____}$$

d is the diameter of rope = 0.02m

N is the speed of engine in rpm

Mass of fuel consumed

$$mfc = \frac{0.82 \times 3600 \times X}{1000 t} \quad (kg/hr)$$

Where,

X is the burette reading in cc

0.82 is the density of diesel in gram per cc

t is the time taken in seconds

Specific fuel consumption

$$sfc = \frac{mfc}{BP} \quad (kg/kW hr)$$

Actual volume of air sucked into the cylinder

$$V_a = C_d A \sqrt{2 g H} 3600 \quad (m^3/hr)$$

Where,

H is the head in meter of air

$$H = \frac{h \rho_w}{1000 \rho_a}$$

A is the area of orifice

h is the manometer reading in mm

ρ_w is the density of water = 1000kg/m³

ρ_a is the density of air = 1.193kg/m³

C_d is the coefficient of discharge = 0.62

Swept volume

$$V_s = \left(\frac{\pi d^2}{4} L \frac{N}{2} 60 \right)$$

Where,

d is the diameter of bore = 80 mm

L is the length of stroke = 110 mm

N is the speed of engine in rpm

Volumetric efficiency

$$\eta_v = \frac{V_A}{V_s} 100 \quad (\%)$$

Brake thermal efficiency

$$\eta_{bth} = \frac{BP \times 3600 \times 100}{mfc \times CV}$$

Where,

CV is the calorific value of diesel = 45200kJ/kg

BP is the brake power in kW

Mechanical efficiency

$$\eta_{mech} = \frac{BP}{IP}$$

Where,

IP is the indicated power in kW

Experimentation:**Aim:**

The experiment is conducted to

- 1) To study and understand the performance characteristics of the engine.
- 2) To draw Performance curves and compare with standards.

Graphs to be plotted:

- 1) SFC v/s BP
- 2) η_{bth} v/s BP
- 3) η_{mech} v/s BP
- 4) η_{vol} v/s BP

Precautions:

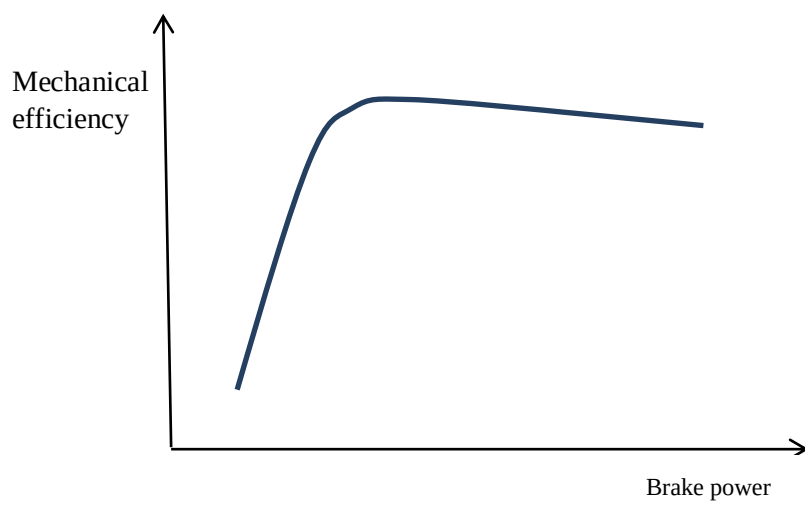
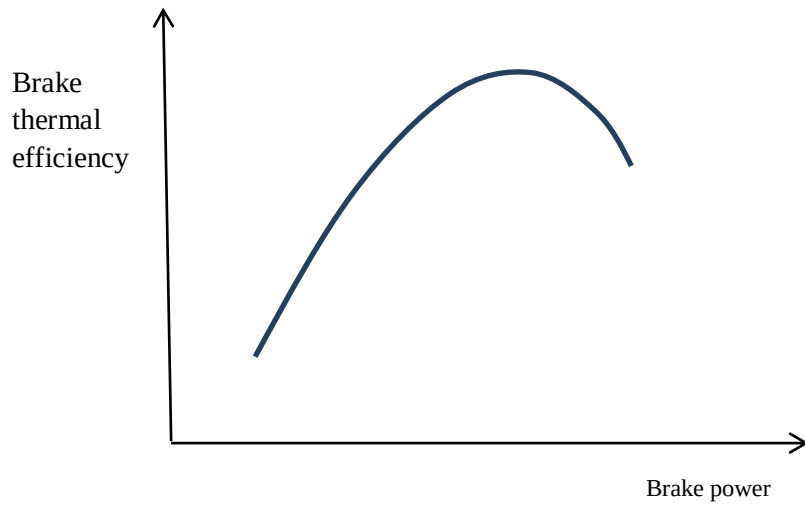
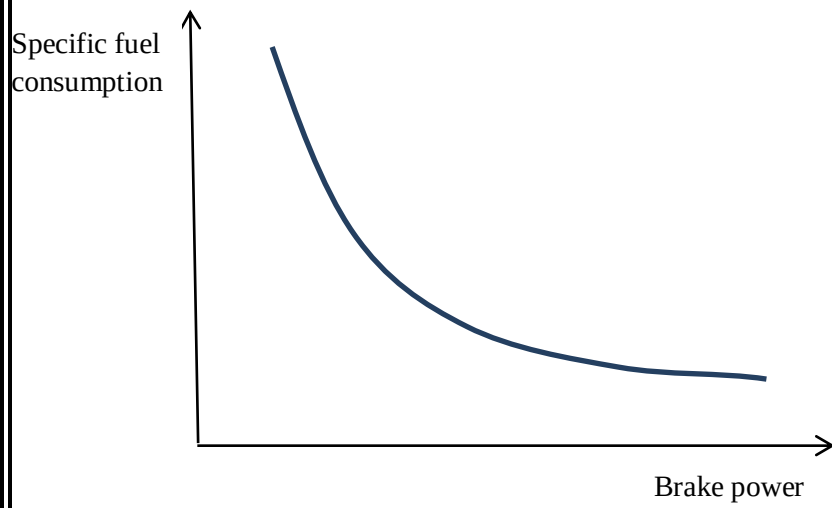
1. Do not run the engine if supply voltage is less than 180V
2. Do not run the engine without the supply of water.
3. Supply water free from dust to prevent blockage in rotameters, engine head and calorimeter.

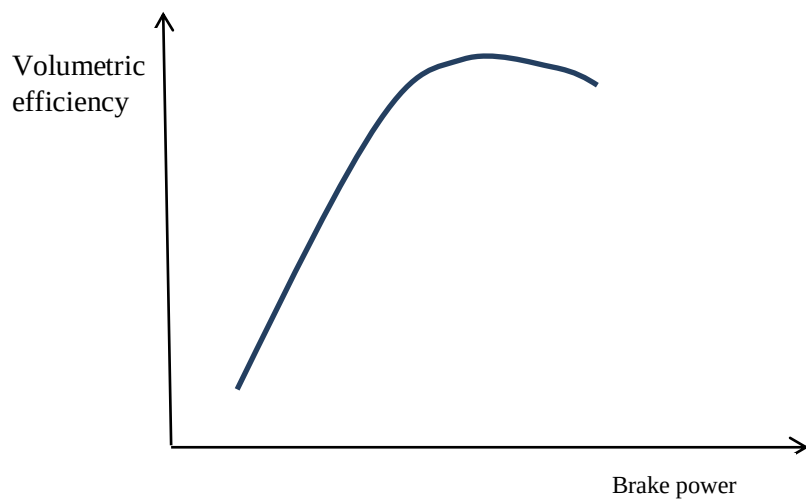
4. Note that the range for water supply provided is an approximate standard values, however the user may select the operating range to his convenience not less than 3 & 2 LPM for engine and calorimeter respectively.
5. Do not forget to give electrical earth and neutral connections correctly.
6. It is recommended to run the engine at 1500rpm otherwise the rotating parts and bearing of engine may run out.

Procedure:

- a. Give the necessary electrical connections to the panel.
- b. Check the lubricating oil level in the engine.
- c. Check the fuel level in the tank.
- d. Allow the water to flow to the engine and the calorimeter and adjust the flow rate.
- e. Release the load if any on the dynamometer.
- f. Open the three-way cock so that fuel flows to the engine.
- g. Start the engine by cranking.
- h. Allow to attain the steady state.
- i. Load the engine by slowly tightening the yoke rod handle of the Rope brake drum.
- j. Note the following readings for particular condition,
 - i. Engine Speed
 - ii. Time taken for ____cc of diesel consumption
 - iii. Rotameter reading.
 - iv. Manometer readings, in cm of water &
 - v. Temperatures at different locations.
- k. Repeat the experiment for different loads and note down the above readings.
- l. After the completion release the load and then switch off the engine.
- m. Allow the water to flow for few minutes and then turn it off.

Expected curve:





Result:

2. PERFORMANCE TEST ON FOUR CYLINDER 4 STROKE DIESEL ENGINE

Introduction

A machine, which uses heat energy obtained from combustion of fuel and converts it into mechanical energy, is known as a Heat Engine. They are classified as External and Internal Combustion Engine. In an External Combustion Engine, combustion takes place outside the cylinder and the heat generated from the combustion of the fuel is transferred to the working fluid which is then expanded to develop the power. An Internal Combustion Engine is one where combustion of the fuel takes place inside the cylinder and converts heat energy into mechanical energy. IC engines may be classified based on the working cycle, thermodynamic cycle, speed, fuel, cooling, method of ignition, mounting of engine cylinder and application.

Diesel Engine is an internal combustion engine, which uses heavy oil or diesel oil as a fuel and operates on two or four stroke. In a 4-stroke Diesel engine, the working cycle takes place in two revolutions of the crankshaft or 4 strokes of the piston. In this engine, pure air is sucked to the engine and the fuel is injected with the combustion taking place at the end of the compression stroke. The power developed and the performance of the engine depends on the condition of operation. So it is necessary to test an engine for different conditions based on the requirement.

Specification

Engine	: Four stroke four cylinder water cooled diesel engine
Make	: Ambassador classic
Power	: 40 BHP @ 4000 rpm
Torque	: 8.6 kg. m @ 2000 rpm
Fuel	: Diesel
Bore	: 73 mm
Stroke length	: 88.9 mm

Starting : Self start

Compression ratio : 23:1

Displacement volume : 1489 cc

Method of ignition : Compression ignition

Orifice diameter : 25 mm

DESCRIPTION OF THE APPARATUS:

The test rig is built for loading mentioned below:

1. The equipment consists of Ambassador Diesel Engine (self-started) of 40 BHP capacity and is Water cooled. The Engine is coupled to an AC generator for loading purposes. Coupling is done by an extension shaft in a separate bearing house.
2. Thermocouples are provided at appropriate positions and are read by a digital temperature indicator with channel selector to select the position.
3. Rota meters are used for direct measurement of water flow rate to the engine and calorimeter respectively.
4. Engine Speed at various conditions is determined by a Digital RPM.
5. A separate air box with orifice assembly is provided for regularizing and measuring the flow rate of air. The pressure difference at the orifice is measured by means of Manometer.
6. A volumetric flask with a fuel distributor is provided for measurement and directing the fuel to the engine respectively.
7. The engine test rig is directly coupled to an AC generator, 3 phase, 10kW, 50Hz, which in turn is loaded by resistance load bank. The load can be varied in steps of $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ and full load by operating the rotary switches provided in the load bank panel

Equations used:

Brake power

$$BP = \sqrt{3} VI \cos \phi \quad (\text{in kW})$$

Where, V is the voltage across two phase

I is the current in each phase

ϕ is the power factor

Calculations:

Brake power

$$BP = \sqrt{3} VI \cos \phi = \text{_____} kW$$

Mass of fuel consumed

$$mfc = \frac{0.82 \times 3600 \times X}{1000 t} = \text{_____} kg/hr$$

Specific fuel consumption

$$sfc = \frac{mfc}{BP} = \text{_____} kg/kW hr$$

Actual volume of air sucked into the cylinder

$$V_a = C_d A \sqrt{2 g H} 3600 = \text{_____} m^3/hr$$

$$H = \frac{h \rho_{Hg}}{1000 \rho_a} = \text{_____}$$

Swept volume

$$V_s = \left(\frac{\pi d^2}{4} L \frac{N}{2} 60 \right) 3 = \text{_____}$$

Volumetric efficiency

$$\eta_v = \frac{V_a}{V_s} 100 = \text{_____} \%$$

Brake thermal efficiency

$$\eta_{bth} = \frac{BP 3600 \times 100}{mfc \times CV} = \text{_____}$$

$$CV = 42500 \text{ kJ/kg}$$

Mechanical efficiency

$$\eta_{mech} = \frac{BP}{IP} = \text{_____}$$

Mass of fuel consumed

$$mfc = \frac{0.82 \times 3600 \times X}{1000 t} \quad (kg/hr)$$

Where,

X is the burette reading in cc

0.82 is the density of diesel in gram per cc

t is the time taken in seconds

Specific fuel consumption

$$sfc = \frac{mfc}{BP} \quad (kg/kW hr)$$

Actual volume of air sucked into the cylinder

$$V_a = C_d A \sqrt{2 g H} 3600 \quad (m^3/hr)$$

Where,

H is the head in meter of air

$$H = \frac{h \rho_{Hg}}{1000 \rho_a}$$

A is the area of orifice

h is the manometer reading in mm

ρ_w is the density of water = 1000kg/m³

ρ_a is the density of air = 1.193kg/m³

C_d is the coefficient of discharge = 0.62

Swept volume

$$V_s = \left(\frac{\pi d^2}{4} L \frac{N}{2} 60 \right) 4$$

Where,

d is the diameter of bore = 73 mm

L is the length of stroke = 88.9 mm

N is the speed of engine in rpm

Volumetric efficiency

$$\eta_v = \frac{V_A}{V_s} 100 \quad (\%)$$

Brake thermal efficiency

$$\eta_{bth} = \frac{BP \times 3600 \times 100}{mfc \times CV}$$

Where,

CV is the calorific value of diesel = 45200kJ/kg

BP is the brake power in kW

Mechanical efficiency

$$\eta_{mech} = \frac{BP}{IP}$$

Where,

IP is the indicated power in kW

Experimentation:**Aim:**

The experiment is conducted to

- a) To study and understand the performance characteristics of the engine.
- b) To draw Performance curves and compare with standards.

Graphs to be plotted:

1. SFC v/s BP
2. η_{bth} v/s BP
3. η_{mech} v/s BP
4. η_{vol} v/s BP

Precautions:

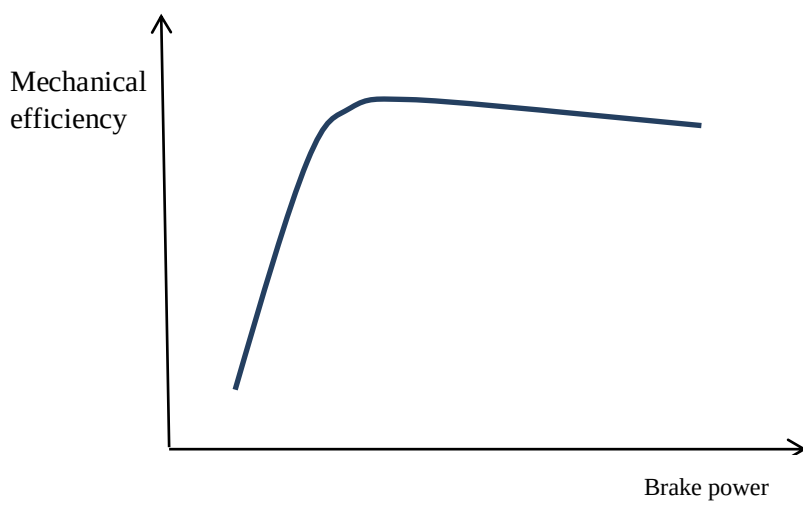
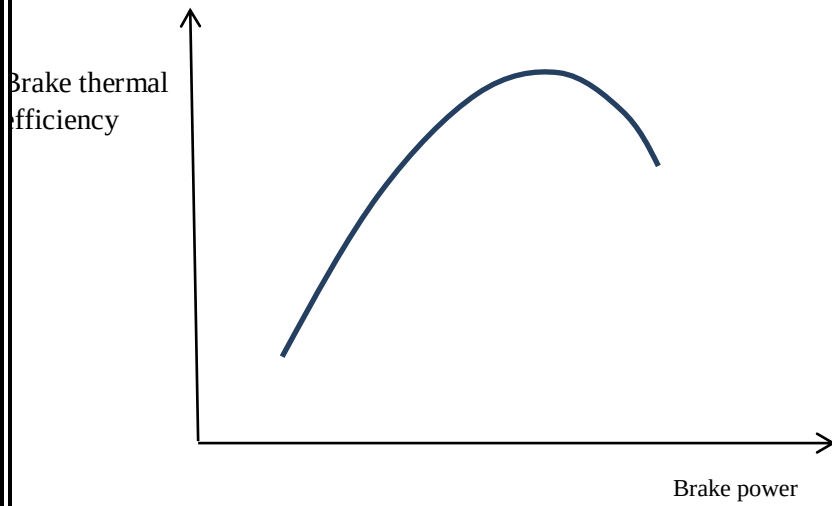
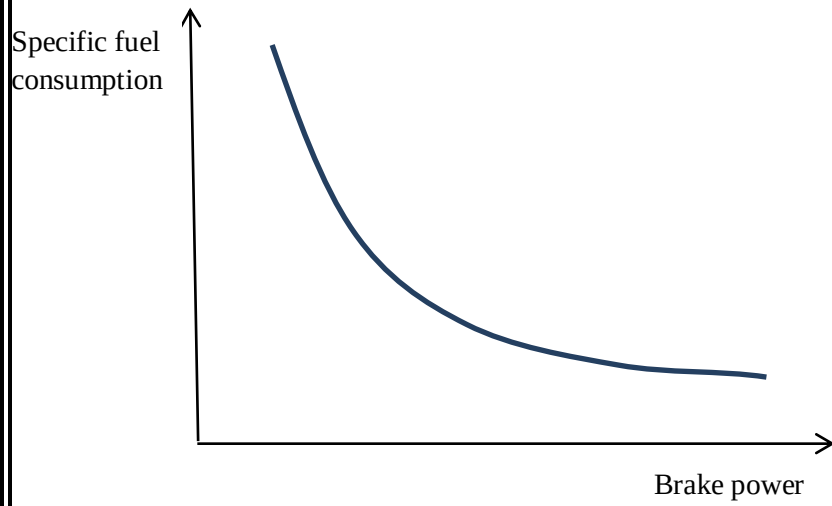
1. Do not run the engine if supply voltage is less than 180V
2. Do not run the engine without the supply of water.
3. Supply water free from dust to prevent blockage in rotameters, engine head and calorimeter.
4. Note that the range for water supply provided is an approximate standard values, however the user may select the operating range to his convenience not less than 3 & 2 LPM for engine and calorimeter respectively.

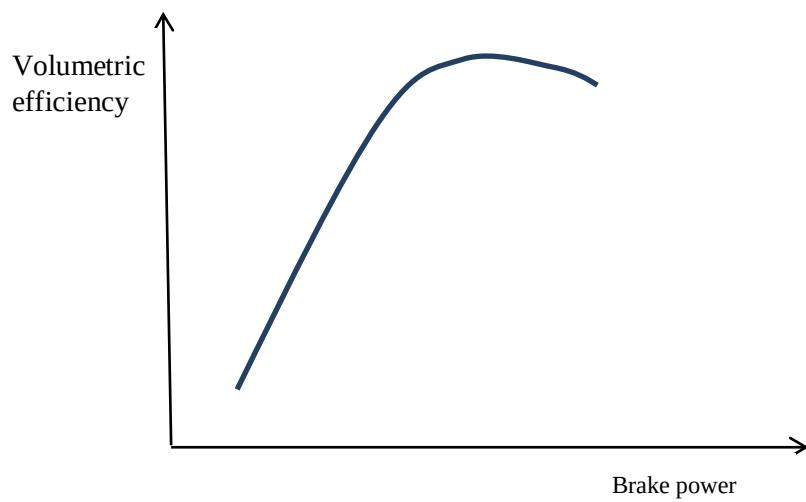
5. Do not forget to give electrical earth and neutral connections correctly.
6. It is recommended to run the engine at 1500rpm otherwise the rotating parts and bearing of engine may run out.

Procedure:

- a. Give the necessary electrical connections to the panel.
- b. Check the lubricating oil level in the engine.
- c. Check the fuel level in the tank.
- d. Allow the water to flow to the engine and the calorimeter and adjust the flow rate.
- e. Release the load if any on the AC generator.
- f. Open the three-way cock so that fuel flows to the engine.
- g. Start the engine.
- h. Allow to attain the steady state.
- i. Load the engine by operating the rotary switches on the resistance of the load bank, in steps of $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, and full load.
- j. Note the following readings for particular condition,
 - i. Engine Speed
 - ii. Time taken for ____cc of diesel consumption
 - iii. Rotameter reading.
 - iv. Manometer readings, in cm of water &
- k. Repeat the experiment for different loads and note down the above readings.
- l. After the completion release the load and then switch off the engine.
- m. Allow the water to flow for few minutes and then turn it off.

Expected curve:





Result:

3. LOAD TEST ON 3 - CYLINDER 4 - STROKE PETROL ENGINE

Introduction

A machine, which uses heat energy obtained from combustion of fuel and converts it into mechanical energy, is known as a Heat Engine. They are classified as External and Internal Combustion Engine. In an External Combustion Engine, combustion takes place outside the cylinder and the heat generated from the combustion of the fuel is transferred to the working fluid which is then expanded to develop the power. An Internal Combustion Engine is one where combustion of the fuel takes place inside the cylinder and converts heat energy into mechanical energy. IC engines may be classified based on the working cycle, thermodynamic cycle, speed, fuel, cooling, method of ignition, mounting of engine cylinder and application.

Description of the apparatus:

a)Hydraulic Dynamometer Loading

- 1) The equipment consists of a Maruti 800 Engine (Self started) of capacity 800cc and power 12HP.
- 2) The Engine is coupled to a Hydraulic Dynamometer for loading purposes. The coupling is done by a universal coupling in a bearing house.
- 3) Thermocouples are provided at appropriate positions and are read by a digital temperature indicator with channel selector.
- 4) Rotameters are used for direct measurement of water flow rate to the engine and calorimeter respectively.
- 5) The Engine Speed and the load applied at various conditions is determined by a Digital RPM Indicator and Load cell arrangement.
- 6) A separate air box with orifice assembly is provided for regularizing and measuring the flow rate of air. The pressure difference is measured by means of a Manometer.

Observations and calculations:

[illegible]

7) A burette with a fuel distributor is provided for measurement and directing the fuel to the engine respectively.

Engine specification

Engine	: Maruti 800
BHP	: 12HP
Speed	: 2000rpm
Fuel	: Petrol
No. of cylinders	: 3
Bore	: 68.5mm
Stroke length	: 72mm
Starting	: Self start
Working cycle	: four stroke
Method of cooling	: water cooled
Method of ignition	: spark ignition

Instrumentation

Digital RPM indicator to measure the speed of the engine.

Digital temperature indicator to measure temperatures at various points.

Differential manometer to measure quantity of air sucked into cylinder.

Burette with stop cock to measure the rate of fuel consumed.

Spring balance to measure load applied to the engine.

b) Fuel measurement

The fuel is supplied to the engine from the main fuel tank through a graduated measuring fuel gauge (Burette) with 2-way cock. To measure the fuel consumption of the engine, fill the burette by opening the cock. By starting a stop clock, measure the time taken to consume 10cc of fuel by the engine.

c) Air intake measurement

The suction side of the engine is connected to an Air tank. The atmospheric air is drawn into the engine cylinder through the air tank. The manometer is provided to measure the pressure drop across an orifice provided in the intake pipe of an air tank. This pressure drop is used to calculate the volume of air drawn into the cylinder. (Orifice diameter is 25mm)

Calculations:

Brake power

$$BP = \frac{W \cdot N}{C} = \text{_____} kW$$

Mass of fuel consumed

$$mfc = \frac{0.72 \times 3600 \times X}{1000 t} = \text{_____} kg/hr$$

Specific fuel consumption

$$sfc = \frac{mfc}{BP} = \text{_____} kg/kW hr$$

Actual volume of air sucked into the cylinder

$$V_a = C_d A \sqrt{2 g H} 3600 = \text{_____} m^3/hr$$

$$H = \frac{h \rho_w}{1000 \rho_a} = \text{_____} mm$$

$$h = H_1 - H_2 = \text{_____} mm$$

Swept volume

$$V_s = \left(\frac{\pi d^2}{4} L \frac{N}{2} 60 \right) 3 = \text{_____}$$

Volumetric efficiency

$$\eta_v = \frac{V_A}{V_s} 100 = \text{_____} \%$$

Brake thermal efficiency

$$\eta_{bth} = \frac{BP 3600 \times 100}{mfc \times CV} = \text{_____}$$

$$CV = 43500 kJ/kg$$

Mechanical efficiency

$$\eta_{mech} = \frac{BP}{IP} = \text{_____}$$

d) Lubrication

The engine is lubricated by mechanical lubrication.

Lubricating oil recommended – SAE – 40 or equivalent.

e) Temperature measurement

A digital temperature indicator with selector switch is provided on the panel to read the temperature in degree centigrade, directly sensed by respective thermocouples located at different places on the test rig.

Note: use the selector knob gently.

Thermocouple details

T₁ : Inlet water temperature of engine jacket and calorimeter

T₂ : Outlet water temperature of engine jacket

T₃ : Outlet water temperature from the calorimeter

T₄ : Exhaust gas temperature from calorimeter

f) Water flow measurement

A flowmeter is provided at the inlet jacket to measure the quantity of water allowed into the engine jacket. A rotameter is provided at inlet of calorimeter. Valves are provided to regulate the flow rate of water. Readings can be taken from flow meter in litres and in rotameter in cc/sec.

g) Loading system

The engine is loaded by varying the quantity of water allowed into the dynamometer at constant pressure head. We can vary the load by operating the gate valve provided on the inlet line of the dynamometer. The dynamometer outlet at the bottom can be adjusted depending upon the load conditions.

Experimentation:

Aim:

The experiment is conducted to

- a. To study and understand the performance characteristics of the engine.
- b. To draw Performance curves and compare with standards.

Equations used:

Brake power

$$BP = \frac{W \cdot N}{C} \quad (\text{in kW})$$

Where,

W is the spring balance reading in kg

N is the speed of engine in rpm

C is a constant, $C = 3240$

Mass of fuel consumed

$$mfc = \frac{0.72 \times 3600 \times X}{1000 t} \quad (kg/hr)$$

Where,

X is the burette reading in cc

0.72 is the density of petrol in gram per cc

t is the time taken in seconds

Specific fuel consumption

$$sfc = \frac{mfc}{BP} \quad (kg/kW hr)$$

Actual volume of air sucked into the cylinder

$$V_a = C_d A \sqrt{2 g H} 3600 \quad (m^3/hr)$$

Where,

H is the head in meter of air

$$H = \frac{h \rho_w}{1000 \rho_a}$$

A is the area of orifice

h is the manometer reading in mm

$$h = H_1 - H_2$$

H₁ and H₂ are the manometer readings in mm

ρ_w is the density of water = 1000kg/m³

ρ_a is the density of air = 1.193kg/m³

C_d is the coefficient of discharge = 0.62

Swept volume

$$V_s = \left(\frac{\pi d^2}{4} L \frac{N}{2} 60 \right) 3$$

Where,

d is the diameter of bore = 68.5mm

L is the length of stroke = 72mm

N is the speed of engine in rpm

Volumetric efficiency

$$\eta_v = \frac{V_A}{V_s} 100 \quad (\%)$$

Brake thermal efficiency

$$\eta_{bth} = \frac{BP \ 3600 \times 100}{mfc \times CV}$$

Where,

CV is the calorific value of petrol = 43500kJ/kg

BP is the brake power in kW

Mechanical efficiency

$$\eta_{mech} = \frac{BP}{IP}$$

Where,

IP is the indicated power in kW

Graphs to be plotted:

1. SFC v/s BP
2. η_{bth} v/s BP
3. η_{mech} v/s BP
4. η_{vol} v/s BP

Procedure:

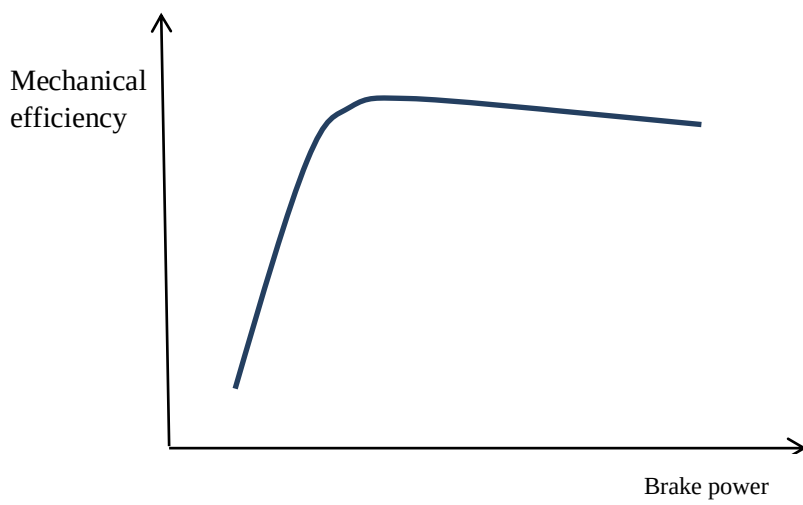
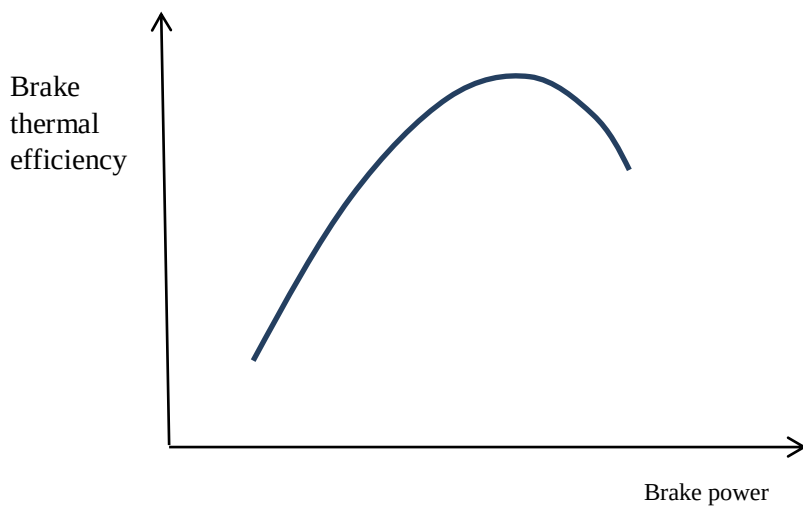
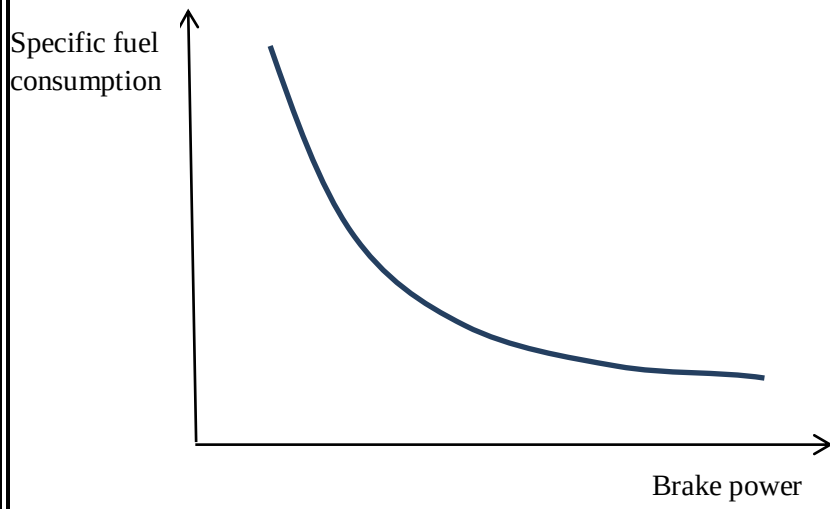
1. Check the lubricating oil level.
2. Check the fuel level.
3. Check and Release the load on the dynamometer if loaded.
4. Check the necessary electrical connections and switch on the panel.
5. Provide the Battery Connections.
6. Allow cooling water to engine and calorimeter.
7. Open the three-way cock so that fuel flows to the engine.
8. Start the engine using the starter key.
9. Set the speed of the engine. (Do not exceed 2000rpm)
10. #Now slowly load the engine using the loading wheel of the dynamometer.

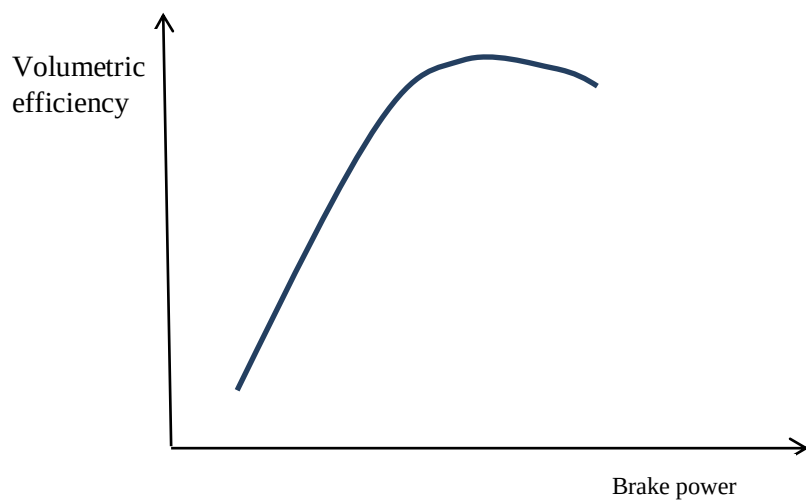
11. Set the engine speed to before rating.
 12. Note the following readings.
 - a. Engine Speed.
 - b. Time taken for 10cc of petrol consumption
 - c. Rotameter readings.
 - d. Manometer readings, in 'm' of water &
 - e. Temperatures.
 13. #Repeat the experiment for other loadings.
 14. After the completion release the load (while doing so release the accelerator) and then switch of the engine and the panel.
 15. *Allow the water to flow for few minutes and then turn it off.
- *Note: Allow water only to the calorimeter and not to the engine.
16. Remove earthing connection of the battery and the starter key.
- #Note: Calculate the maximum permissible load on the engine using the equation for rated BHP (given in the specification).

Precautions:

1. Do not run the engine if supply voltage is less than 180V
2. Do not run the engine without the supply of water.
3. Supply water free from dust to prevent blockage in rotameters, engine head and calorimeter.
4. Note that the range for water supply provided is an approximate standard values, however the user may select the operating range to his convenience not less than 3 & 2 LPM for engine and calorimeter respectively.
5. Always set the accelerator knob to the minimum condition and start the engine.
6. Do not forget to give electrical earth and neutral connections correctly.
7. It is recommended to run the engine below 3000rpm otherwise the rotating parts and bearing of dynamometer may run out.

Expected curve:





Result:

4.Air Blower Test Rig

Introduction

The blower is single stage centrifugal type. The air is sucked from atmosphere from suction side. The slightly compressed air passes through the spiral case, before it comes out through the outlet.

Blowers are used to discharge higher volumes of air at low pressures. These are used in blast furnaces, cupolas, mines, air conditioning plants, etc. The three types of impellers namely: Backward curved (red), Radial (yellow), Forward curved (black) can be fixed internally at a time.

The blower consists of an impeller fabricated to the shapes. The vanes of the impeller is pressed out of the sheet metal and riveted to the two shrouds thus forming into a unique unit. The other impellers are also fabricated in the same fashion and are easily mountable on the shaft. The casing is made of cast iron. The whole contour is volute shape. The smooth contour helps in reducing eddy current losses along the path. The casing is designed in such a way that it can be separated to facilitate mounting or interchanging of impellers.

The shaft is coupled to an induction motor with the help of a belt and having the facility to change three different speeds with the help of a stepped cone pulley. The input power of the motor can be measured by an energy meter provided on the panel.

The flow of air at inlet and outlet is measured by the manometers. The whole probe can be actuated by the screw mechanism. This render to draw profile of the velocity along the diameter of the pipe.

The pitot tube is used to find out the dynamic and static pressure at suction and delivery side. Venturi is used to find out the quantity of air flow rate, with help of water manometer.

Specification

Type	: Centrifugal type
Motor	: 3 HP Induction motor
Speed (max)	: 2800 rpm
Type of impellers	: Backward curved (red)
	: Radial (yellow)
	: Forward curved (black)

Observations:

Sl no.	Position of valve	Pitot tube 1 (cm)			Venturimeter readings (cm)			Time for 5 revolution (sec)	Speed (rpm)
		h ₁	h ₂	h _{sta}	h ₁	h ₂	h _f		
1	¼ open								
2	½ open								
3	¾ open								
4	Full open								

Calculations:

Input power to motor

$$IP_m = \frac{n \ 3600}{t \ E} = \text{_____} kW$$

E is the energy meter constant = 200 rev/ kW hr

Input power to blower

$$IP_b = IP_m \eta_t = \text{_____} kW$$

Static head developed due to air (Pitot tube 1)

$$H_{sta} = \frac{h_{sta} \ \rho_w}{1000 \ \rho_a} = \text{_____} m \ of \ air$$

ρ_w is the density of water = 1000 kg/m³

ρ_a is the density of air = 1.193 kg/m³

Volume flow rate

$$Q = C_d \ K \ \sqrt{2 \ g \ H_f} = \text{_____} m^3/sec$$

C_d is the coefficient of discharge of venturimeter = 0.92

g is the acceleration due to gravity = 9.81 m/sec²

$$K = \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}} = \text{_____} m^2$$

Equations used:

Input power to the motor

$$IP_m = \frac{n \ 3600}{t \ E} \quad (kW)$$

Where,

n is the number of revolutions

t is the time taken in seconds for the five revolutions of the energy meter

E is the energy meter constant = 200 rev/ kW hr

Input power to the blower

Input power to blower

$$IP_b = IP_m \eta_t \ (kW)$$

Where,

η_t is the transmission efficiency = 90%

Static head developed due to air (Pitot tube)

$$H_{sta} = \frac{h_{sta} \rho_w}{1000 \rho_a} \quad (m \ of \ air)$$

Where,

h_{sta} is the manometer reading from pitot tube at the entry in mm

ρ_w is the density of water = 1000 kg/m³

ρ_a is the density of air = 1.193 kg/m³

Volume flow rate

$$Q = C_d K \sqrt{2 g H_f} \quad (m^3/sec)$$

Where,

C_d is the coefficient of discharge of venturimeter = 0.92

g is the acceleration due to gravity = 9.81 m/sec²

$$K = \frac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}}$$

Area at the entry of venturimeter

$$A_1 = \frac{\pi d_1^2}{4} \quad (m^2)$$

Area at the entry of venturimeter

$$A_1 = \frac{\pi d_1^2}{4} = \text{_____} m^2$$

d_1 is the diameter at the entry of venturimeter = 0.15 m

Area at the throat of venturimeter

$$A_2 = \frac{\pi d_2^2}{4} = \text{_____} m^2$$

d_2 is the diameter at the throat of venturimeter = 0.1 m

Pressure head

$$H_f = \frac{h_f \rho_w}{1000 \rho_a} = \text{_____} m \text{ of air}$$

Blower output

$$P = \frac{\rho_a g Q H_{total}}{1000} = \text{_____} kW$$

$$H_{total} = H_f + H_{sta} = \text{_____} m \text{ of air}$$

Blower efficiency

$$\eta_{blower} = \frac{P}{IP_b} = \text{_____} \%$$

Overall efficiency

$$\eta_{overall} = \frac{P}{IP_m} = \text{_____} \%$$

Calculation table

Sl No.	Blower input power IP (kW)	Blower output power P (kW)	Volume flow rate Q (m ³ /sec)	Blower efficiency η_{blower}	Overall efficiency $\eta_{overall}$

d_1 is the diameter at the entry of venturimeter = 0.15 m

Area at the throat of venturimeter

$$A_2 = \frac{\pi d_2^2}{4} \quad (m^2)$$

d_2 is the diameter at the throat of venturimeter = 0.1 m

Pressure head

$$H_f = \frac{h_f \rho_w}{1000 \rho_a} \quad (m \text{ of air})$$

h_f is the velocity head in mm read from manometer-3 (venturimeter)

Blower output

$$P = \frac{\rho_a g Q H_{total}}{1000} \quad (kW)$$

Where,

$$H_{total} = H_f + H_{sta}$$

Blower efficiency

$$\eta_{blower} = \frac{P}{IP_b}$$

Overall efficiency

$$\eta_{overall} = \frac{P}{IP_m}$$

Aim

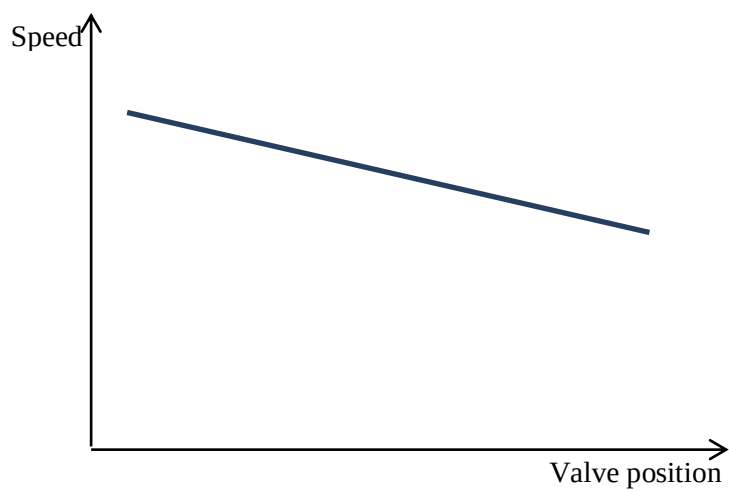
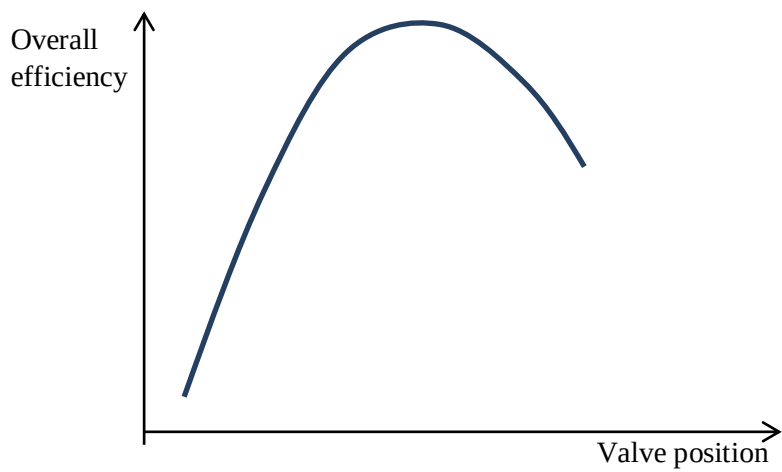
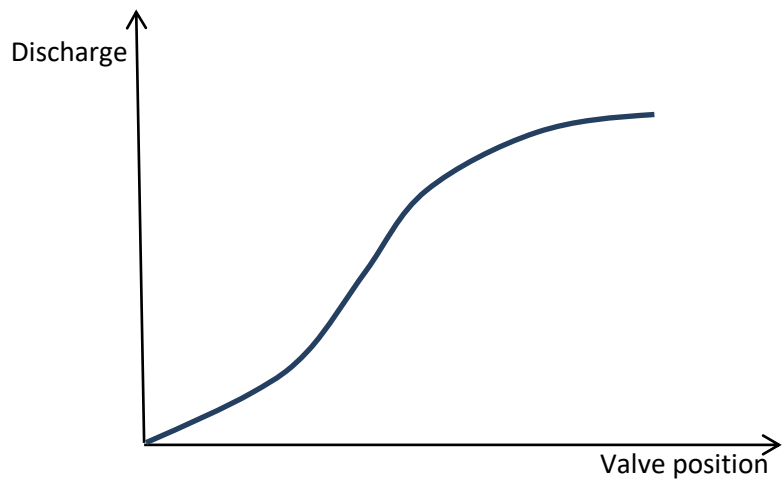
To conduct a performance test on an air blower to determine the overall efficiency.

Plot the graph for different volume discharge v/s overall efficiency.

Procedure

1. Close the gate valve of the blower provided.
2. Start the motor using starter.
3. Now open the gate valve for the desired flow.
4. Note down the venture, pitot tubes and pressure drop readings.
5. Measure the speed of the motor using tacho meter.
6. Measure the input power consumed by the motor using energy meter corresponding to the time.
7. At various stages/positions of gate valve, note down the corresponding readings.
8. Repeat the above procedure for three different speed and also for three different impellers.

Expected curves:



Result:

I. SINGLE STAGE AIR COMPRESSOR TEST RIG

Introduction

A COMPRESSOR is a device, which sucks in air at atmospheric pressure & increases its pressure by compressing it. If the air is compressed in a single cylinder it is called as a Single Stage Compressor. If the air is compressed in two or more cylinders it is called as a Multi Stage Compressor.

In a Two Stage Compressor the air is sucked from atmosphere & compressed in the first cylinder called the low-pressure cylinder. The compressed air then passes through an inter cooler where its temperature is reduced. The air is then passed into the second cylinder where it is further compressed. The air further goes to the air reservoir where it is stored.

Specification

Make	: Bakgiam
Type	: Reciprocating type
Stage	: Single stage
Cylinder	: Single cylinder
Bore	: 100mm
Stroke length	: 80mm
Motor type	: Squirrel cage induction motor
Motor rating	: 5 HP
Motor speed	: 1420 rpm
Compressor speed	: 925 rpm
Electrical supply	: 415 V/380 V, 3 phase, 50Hz
Type of starter	: DOL
Belt size	: B 68
Type of lubrication	: Splash
Type of cooling	: Air cooled
Type of fan	: Forced draught
Diameter of orifice	: 20mm

Equations used:

1. Air head causing flow, h_a

$$h_a = \frac{h_w \rho_{water}}{\rho_a} \quad (m \text{ of air})$$

Observations:

Sl. No.	Compressor Speed, N_c (rpm)	Delivery pressure (bar) absolute	Inlet pressure of compressor P_1 (bar) absolute	Outlet pressure of compressor P_2 (bar) Absolute	Motor speed, N_m (rpm)	T_1 (K)	T_2 (K)	Time for 'n' pulses/revolutions of energy meter, T (sec)	Manometer reading (m)	
									h_1	h_2
1										
2										
3										
4										
5										

Calculations:

1. Air head causing flow, h_a

$$h_a = \frac{(h_1 - h_2)\rho_{water}}{\rho_a} = \text{_____ } m \text{ of air}$$

2. Actual vol. of air compressed, Q_a

$$Q_a = c_d a \sqrt{2 g h_a} = \text{_____ } m^3/s$$

3. Theoretical volume of air compressed, Q_{th}

$$Q_{th} = \frac{\frac{\pi}{4} D^2 L N_c}{60} = \text{_____ } m^3/s$$

4. Input Power to the motor, IP

$$IP = \frac{n \times 3600}{K T} = \text{_____ } kW$$

5. Isothermal Work done, WD

$$WD = \rho_a Q_a R_a T_a \ln R = \text{_____ } W$$

6. Compression ratio

$$R = \frac{P_2}{P_1} = \text{———}$$

Where,

$h_w = (h_1 - h_2)$ is water column reading in m of water

ρ_{water} is density of the water = 1000 kg/m³

ρ_{air} is the density of the air = 1.293 kg/m³

2. Actual vol. of air compressed, Q_a

$$Q_a = c_d a \sqrt{2 g h_a} \quad (m^3/s)$$

Where,

h_a is air head causing the flow in m of air.

C_d is the co-efficient of discharge of orifice = 0.62

a is the area of orifice = $(\pi d^2) / 4$

Where, d = diameter of orifice = 0.02 m

3. Theoretical volume of air compressed, Q_{th}

$$Q_{th} = \frac{\frac{\pi}{4} D^2 L N_c}{60} \quad (m^3/s)$$

Where,

D is the diameter of the cylinder = 0.10 m.

L is Stroke Length = 0.080 m

N_c is speed of the compressor in rpm

4. Input Power to the motor, IP

$$IP = \frac{n \times 3600}{K T} \quad (kW)$$

Where, n = No. of revolutions of energy meter (Say 5)

K = Energy meter constant, revs/kW-hr

T = time for 5 rev. of energy meter in seconds

7. Volumetric efficiency, η_{vol}

$$\eta_{vol} = \frac{Q_a}{Q_{th}} 100 = \underline{\hspace{2cm}}$$

8. Output power of motor, BP

$$BP = \frac{2 \pi N_m \cdot s \cdot g \cdot r}{60 \times 1000} = \underline{\hspace{2cm}} kW$$

9. Power of compressor, P_c

$$IP_c = BP \eta_t = \underline{\hspace{2cm}} kW$$

10. Isothermal efficiency, η_{iso}

$$\eta_{iso} = \frac{\text{Isothermal work done}}{IP_c} 100 = \underline{\hspace{2cm}}$$

11. Overall efficiency, η_{iso}

$$\eta_{overall} = \frac{\text{Isothermal work done}}{IP} 100 = \underline{\hspace{2cm}}$$

Calculation Table

Sl no.	Actual volume of air compressed Q_a (in m^3/s)	Theoretical volume of air compressed Q_{th} (in m^3/s)	Isothermal work done (kW)	Power of compressor P_c (kW)	Output power of motor BP (kW)	Isothermal efficiency η_{iso}	Volumetric efficiency η_{vol}	Overall efficiency $\eta_{overall}$
1								
2								
3								
4								
5								
6								

5. Isothermal Work done, WD

$$WD = \rho_a Q_a R_a T_a \ln R \quad (W)$$

Where,

ρ_a = Density of the air = 1.293 kg/m³

Q_a = Actual volume of air compressed.

R_a = characteristic gas constant = 0.287 kJ/kg K

T_a = Temperature T_1 in kelvin

R = Compression ratio

$$R = \frac{P_2}{P_1}$$

Where,

P_1 = Suction pressure in bar

P_2 = Delivery pressure in bar

NOTE: To convert delivery pressure from kg/cm² to kPa multiply by 98.1

6. Volumetric efficiency, η_{vol}

$$\eta_{vol} = \frac{Q_a}{Q_{th}} 100$$

7. Isothermal efficiency, η_{iso}

$$\eta_{iso} = \frac{\text{Isothermal work done}}{IP_c} 100$$

8. Overall efficiency, η_{iso}

$$\eta_{iso} = \frac{\text{Isothermal work done}}{IP} 100$$

9. Output power of motor, BP

$$BP = \frac{2 \pi N_m s . g . r}{60 \times 1000}$$

Where,

N_m = speed of motor

s = load cell reading in N

r = radius of the arc = 190 mm

g = acceleration due to gravity

10. Power of compressor, P_c

$$P_c = IP \eta_m$$

Where,

η_m = transmission efficiency = 80%

Experimentation:

Aim:

The experiment is conducted at various pressures to

- a. Determine the Volumetric efficiency.
- b. Determine the Isothermal efficiency.

Procedure:

1. Check the lubricating oil level in the compressor.
2. Start the compressor by switching on the motor.
3. The slow increase of the pressure inside the air reservoir is observed.
4. Maintain the required pressure by slowly operating the discharge valve (open/close). (Note there may be slight variations in the pressure readings since it is a dynamic process and the reservoir will be filled continuously till the cut-off.)
5. Now note down the following readings in the respective units,

Speed of the compressor.

Manometer readings.

Inlet air pressure

Discharge pressure.

Temperatures.

Energy meter reading.

6. Repeat the experiment for different delivery pressures.
7. Once the set of readings are taken switch off the compressor.
8. The air stored in the tank is discharged. Be careful while doing so, because the compressed air passing through the small area also acts as a air jet which may damage you or your surroundings.
9. Repeat the above two steps after every experiment.

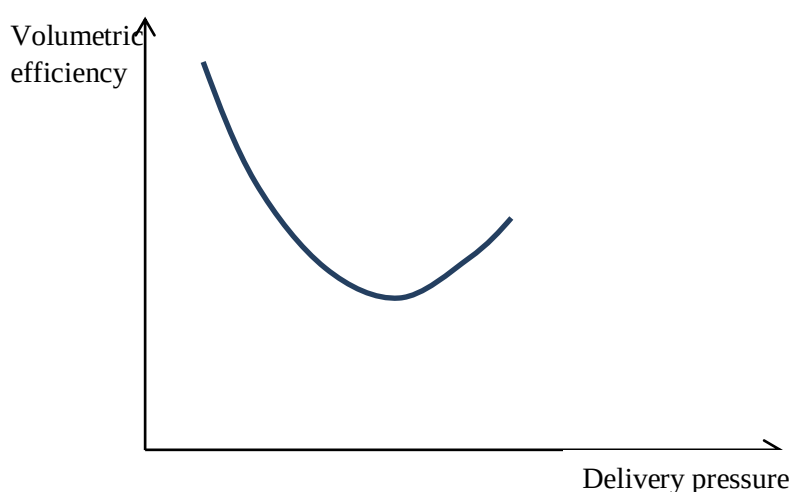
Graphs to be plotted:

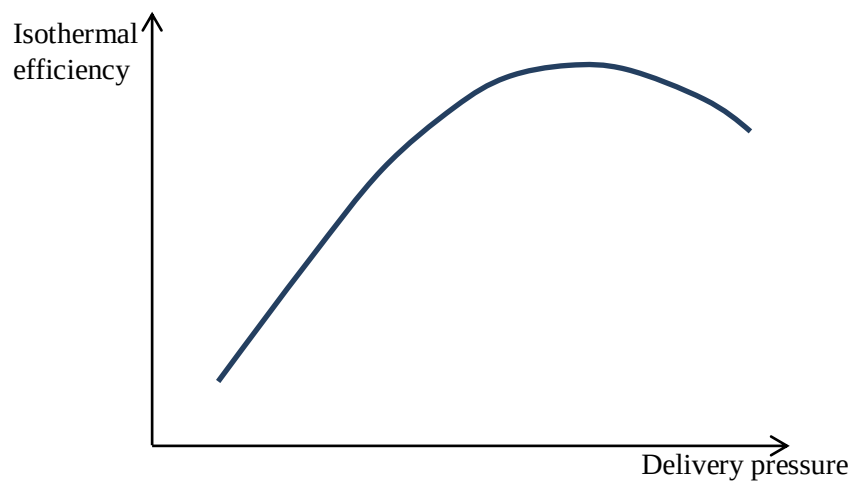
1. Delivery Pressure vs. η_{vol}
2. Delivery Pressure vs. η_{iso}

Precautions:

1. Do not run the blower if supply voltage is less than 380V
2. Check the direction of the motor, if the motor runs in opposite direction change the phase line of the motor to run in appropriate direction.
3. Do not forget to give electrical earth and neutral connections correctly.

Expected curves:





Result:

THERMAL ENGINEERING LAB

HEAT ENGINE LAB

CYCLE – 2

1. PERFORMANCE TEST ON MULTY STAGE RECIPROCATING AIR COMPRESSOR TEST RIG
2. HEAT BALANCE SHEET ON FOUR STROKE SINGLE CYLINDER DIESEL ENGINE
3. HEAT BALANCE SHEET ON FOUR STROKE FOUR CYLINDER DIESEL ENGINE
4. HEAT BALANCE SHEET ON FOUR STROKE THREE CYLINDER PETROL ENGINE
5. MORSE TEST ON THREE CYLINDER FOUR STROKE PETROL ENGINE

LAB IN CHARGE

Professor : Dr. Gireesh Kumaran Thampi

1. MULTI STAGE AIR COMPRESSOR TEST RIG

Introduction

A COMPRESSOR is a device, which sucks in air at atmospheric pressure & increases its pressure by compressing it. If the air is compressed in a single cylinder it is called as a Single Stage Compressor. If the air is compressed in two or more cylinders it is called as a Multi Stage Compressor.

In a Two Stage Compressor the air is sucked from atmosphere & compressed in the first cylinder called the low-pressure cylinder. The compressed air then passes through an inter cooler where its temperature is reduced. The air is then passed into the second cylinder where it is further compressed. The air further goes to the air reservoir where it is stored.

Specification

Make	: Bakgiam
Type	: Reciprocating type
Stage	: Two stage
Cylinder	: Two cylinders
Low pressure bore	: 60.33 mm
High pressure bore	: 57 mm
Stroke length of LP-HP piston	: 62 mm
Motor type	: Squirrel cage induction motor
Motor rating	: 2 HP
Motor speed	: 1440 rpm
Compressor speed	: 740 rpm
Electrical supply	: 415 V/380 V, 3 phase, 50Hz
Type of starter	: DOL
Belt size	: A 68
Type of lubrication	: Splash
Type of cooling	: Air cooled
Type of fan	: Forced draught
Diameter of orifice	: 20mm

Equations used:

1. Air head causing flow, h_a

$$h_a = \frac{h_w \rho_{water}}{\rho_a} \quad (m \text{ of air})$$

Observations:

Sl. No.	Compressor Speed, N_c (rpm)	Inlet pressure of compressor P_1 (bar)	Intermediate pressure of compressor P_2 (bar)	Exit pressure of compressor P_3 (bar)	Motor speed, N_m (rpm)	Time for 'n' revolutions of energy meter, T (sec)	Manometer meter reading (m)	
							h_1	h_2
1								
2								
3								
4								
5								

Sl no.	Temperature readings			
	T_1 (K)	T_2 (K)	T_3 (K)	T_4 (K)

Where,

$h_w = (h_1 - h_2)$ is water column reading in m of water

ρ_{water} is density of the water = 1000 kg/m³

ρ_{air} is the density of the air = 1.293 kg/m³

2. Actual vol. of air compressed, Q_a

$$Q_a = c_d a \sqrt{2 g h_a} \quad (m^3/s)$$

Where,

h_a is air head causing the flow in m of air.

C_d is the co-efficient of discharge of orifice = 0.62

a is the area of orifice = $(\pi d^2) / 4$

Where, d = diameter of orifice = 0.02 m

3. Theoretical volume of air compressed, Q_{th}

$$Q_{th} = \frac{\frac{\pi}{4} D_1^2 L N_c}{60} + \frac{\frac{\pi}{4} D_2^2 L N_c}{60} \quad (m^3/s)$$

Where,

D_1 is the diameter of the low pressure cylinder = 60.33 mm.

D_2 is the diameter of the high pressure cylinder = 57 mm.

L is Stroke Length = 0.062 m

N_c is speed of the compressor in rpm

4. Input Power to the electric motor, IP

$$IP = \frac{n \times 3600}{K T} \quad (kW)$$

Where, n = No. of revolutions of energy meter (Say 5)

K = Energy meter constant, revs/kW-hr

T = time for 5 rev. of energy meter in seconds

Calculations:

1. Air head causing flow, h_a

$$h_a = \frac{(h_1 - h_2)\rho_{water}}{\rho_a} = \text{_____} \text{ m of air}$$

2. Actual vol. of air compressed, Q_a

$$Q_a = c_d a \sqrt{2 g h_a} = \text{_____} \text{ m}^3/\text{s}$$

3. Theoretical volume of air compressed, Q_{th}

$$Q_{th} = \frac{\frac{\pi}{4} D_1^2 L N_c}{60} + \frac{\frac{\pi}{4} D_2^2 L N_c}{60} = \text{_____} \text{ m}^3/\text{s}$$

4. Input Power to the electric motor, IP

$$IP = \frac{n \times 3600}{K T} = \text{_____} \text{ kW}$$

5. Isothermal Work done, WD

$$WD = \rho_a Q_a R_a T_a \ln R = \text{_____} \text{ W}$$

6. Compression ratio

$$R = \frac{P_3}{P_1} = \text{_____}$$

7. Volumetric efficiency, η_{vol}

$$\eta_{vol} = \frac{Q_a}{Q_{th}} 100 = \text{_____}$$

8. Isothermal efficiency, η_{iso}

$$\eta_{iso} = \frac{\text{Isothermal work done}}{IP} 100 = \text{_____}$$

9. Output power of motor, BP

$$BP = \frac{2 \pi N_m s.g.r}{60 \times 1000} = \text{_____} \text{ kW}$$

5. Isothermal Work done, WD

$$WD = \rho_a Q_a R_a T_a \ln R \quad (W)$$

Where,

ρ_a = Density of the air = 1.293 kg/m³

Q_a = Actual volume of air compressed.

R_a = characteristic gas constant = 0.287 kJ/kg K

T_a = Temperature T_1 in kelvin

R = Compression ratio

$$R = \frac{P_3}{P_1}$$

Where,

P_1 = suction pressure in bar

P_3 = delivery pressure in bar

NOTE: To convert delivery pressure from kg/cm² to kPa multiply by 98.1

6. Volumetric efficiency, η_{vol}

$$\eta_{vol} = \frac{Q_a}{Q_{th}} 100$$

7. Isothermal efficiency, η_{iso}

$$\eta_{iso} = \frac{\text{Isothermal work done}}{P_c} 100$$

8. Output power of motor, BP

$$BP = \frac{2 \pi N_m s \cdot g \cdot r}{60 \times 1000}$$

Where,

N_m = speed of motor

s = load cell reading in N

g = acceleration due to gravity

r = radius of the arc = 190 mm

10. Power of compressor, P_c

$$P_c = BP \eta_t = \text{_____} kW$$

Sl no.	Actual volume of air compressed Q_a (in m^3/s)	Theoretical volume of air compressed Q_{th} (in m^3/s)	Isothermal work done (kW)	Power of compressor P_c (kW)	Output power of motor BP (kW)	Isothermal efficiency η_{iso}	Volumetric efficiency η_{vol}
1							
2							
3							
4							
5							
6							

9. Power of compressor, P_c

$$P_c = BP \eta_t \text{ (kW)}$$

Where,

η_t = transmission efficiency = 80%

Experimentation:

Aim:

The experiment is conducted at various pressures to

- a. Determine the Volumetric efficiency.
- b. Determine the Isothermal efficiency.

Procedure:

1. Check the lubricating oil level in the compressor.
2. Start the compressor by switching on the motor.
3. The slow increase of the pressure inside the air reservoir is observed.
4. Maintain the required pressure by slowly operating the discharge valve (open/close). (Note there may be slight variations in the pressure readings since it is a dynamic process and the reservoir will be filled continuously till the cut-off.)
5. Now note down the following readings in the respective units,

Speed of the compressor.

Manometer readings.

Inlet air pressure.

Discharge pressure of first stage

Discharge pressure of first stage

Temperatures.

Energy meter reading.

6. Repeat the experiment for different delivery pressures.
7. Once the set of readings are taken switch of the compressor.
8. The air stored in the tank is discharged. Be careful while doing so, because the compressed air passing through the small area also acts as an air jet which may damage you or your surroundings.
9. Repeat the above two steps after every experiment.

Graphs to be plotted:

- i. Delivery Pressure vs. η_{vol}
- ii. Delivery Pressure vs. η_{iso}

Precautions:

1. Do not run the blower if supply voltage is less than 380V
2. Check the direction of the motor, if the motor runs in opposite direction change the phase line of the motor to run in appropriate direction.
3. Do not forget to give electrical earth and neutral connections correctly.

Result:

2.HEAT BALANCE TEST ON SINGLE CYLINDER 4STROKE DIESEL ENGINE

Introduction

A machine, which uses heat energy obtained from combustion of fuel and converts it into mechanical energy, is known as a Heat Engine. They are classified as External and Internal Combustion Engine. In an External Combustion Engine, combustion takes place outside the cylinder and the heat generated from the combustion of the fuel is transferred to the working fluid which is then expanded to develop the power. An Internal Combustion Engine is one where combustion of the fuel takes place inside the cylinder and converts heat energy into mechanical energy. IC engines may be classified based on the working cycle, thermodynamic cycle, speed, fuel, cooling, method of ignition, mounting of engine cylinder and application.

Diesel Engine is an internal combustion engine, which uses heavy oil or diesel oil as a fuel and operates on two or four stroke. In a 4-stroke Diesel engine, the working cycle takes place in two revolutions of the crankshaft or 4 strokes of the piston. In this engine, pure air is sucked to the engine and the fuel is injected with the combustion taking place at the end of the compression stroke. The power developed and the performance of the engine depends on the condition of operation. So it is necessary to test an engine for different conditions based on the requirement.

Specification

Engine	: Four stroke single cylinder
Make	: Kirloskar
Power	: 5 HP
Engine speed	1500
Fuel	: Diesel
Bore	: 80 mm
Stroke length	: 110 mm

Observations:

Sl no.	Load W-S (kg)	Speed N (rpm)	Flow rate of cooling water to engine jacket, R_{t1} (cc/sec)	Flow rate of cooling water to calorimeter, R_{t2} (cc/sec)

Sl no.	Thermocouple readings (in $^{\circ}\text{C}$)					Time for consuming 10cc of fuel t (sec)
	T_1	T_2	T_3	T_4	T_5	

Calculations:

Brake power

$$BP = \frac{2 \pi N (W - S) \left(\frac{D + d}{2}\right) 9.81}{60000} = \text{_____} \text{ in kW}$$

Mass of fuel consumed

$$mfc = \frac{0.72 \times 3600 \times X}{1000 t} = \text{_____} \text{ kg/hr}$$

Specific fuel consumption

$$sfc = \frac{mfc}{BP} = \text{_____} \text{ kg/kW hr}$$

Heat input

$$Q_{in} = \frac{mfc \times CV}{60} = \text{_____} \text{ kJ/min}$$

Heat equivalent to brake power calculations

$$Q_{bp} = BP \times 60 = \text{_____} \text{ kJ/min}$$

Starting : Cranking

Working cycle : Four stroke

Method of cooling : Water cooled

Method of ignition : Compression ignition

Description of the apparatus:

The test rig is built for loading mentioned below:

1. The equipment consists of KIRLOSKAR Diesel Engine (Crank started) of 5hp (3.7kW) capacity and is Water cooled. The Engine is coupled to a Rope Brake Drum Dynamometer for loading purposes. Coupling is done by an extension shaft in a separate bearing house. The dynamometer is connected to the spring load assembly for varying the load.
2. Thermocouples are provided at appropriate positions and are read by a digital temperature indicator with channel selector to select the position.
3. Rota meters are used for direct measurement of water flow rate to the engine and calorimeter respectively.
4. Engine Speed and the load applied at various conditions is determined by a Digital RPM Indicator and spring balance reading.
5. A separate air box with orifice assembly is provided for regularizing and measuring the flow rate of air. The pressure difference at the orifice is measured by means of Manometer.
6. A volumetric flask with a fuel distributor is provided for measurement and directing the fuel to the engine respectively.

Thermocouple details

T₁ : Inlet water temperature of engine jacket and calorimeter

T₂ : Outlet water temperature of engine jacket

T₃ : Outlet water temperature from the calorimeter

T₄ : Exhaust gas temperature from engine

T₅ : Exhaust gas temperature from the calorimeter

Heat carried away by the engine cooling water

$$Q_w = m_w c_{pw} (T_2 - T_1) = \text{_____} \text{ kJ/min}$$

$$m_w = \rho_w R_{t1} \times 10^{-6} \times 60 = \text{_____} \text{ kg/min}$$

Heat carried away by the exhaust gas

$$Q_e = m_{w1} c_{pw} (T_3 - T_1) \frac{T_4 - T_a}{T_4 - T_5} = \text{_____} \text{ kJ/min}$$

$$c_{pw} = 4.187 \text{ kJ/kgK}$$

$$\rho_w = 1000 \text{ kg/m}^3$$

$$m_{w1} = \rho_w R_{t2} \times 10^{-6} \times 60 = \text{_____} \text{ kg/min}$$

Unaccounted heat or heat loss due to radiation

$$= Q_{in} - (Q_{bp} + Q_w + Q_e) = \text{_____} \text{ kJ/min}$$

Equations used:

Brake power

$$BP = \frac{2 \pi N (W - S) \left(\frac{D + d}{2}\right) 9.81}{60000} \quad (\text{in kW})$$

Where,

S is the spring balance reading in kg

W is the dead weights in kg

D is the diameter of the brake drum = 0.33m

d is the diameter of rope = 0.02m

N is the speed of engine in rpm

Mass of fuel consumed

$$mfc = \frac{0.82 \times 3600 \times X}{1000 t} \quad (\text{kg/hr})$$

Where,

X is the burette reading in cc

0.82 is the density of diesel in gram per cc

t is the time taken in seconds

Specific fuel consumption

$$sfc = \frac{mfc}{BP} \quad (\text{kg/kW hr})$$

Heat input

$$Q_{in} = \frac{mfc \times CV}{60} \quad (\text{kJ/min})$$

CV is the calorific value of fuel = 45200 kJ/kg

Heat equivalent to brake power calculations

$$Q_{bp} = BP \times 60 \quad (\text{kJ/min})$$

Heat balance sheet

Sl. No.	Particulars	Heat Content kW	%
1	Heat Input -- A		100
2	Heat to BP -- B		B/A =
3	Heat to Cooling Water -- C		C/A =
4	Heat to Exhaust Gases -- D		D/A =
5	Heat Unaccounted -- E		E/A =

Heat carried away by the engine cooling water

$$Q_w = m_w c_{pw} (T_2 - T_1) \quad (kJ/min)$$

Where,

c_{pw} is the specific heat of water

ρ_w is the density of water

m_w is the mass flow rate of water into the engine cooling jacket

T_2 is the outlet water temperature of engine water jacket

T_1 is the inlet water temperature of engine water jacket

$$c_{pw} = 4.187 \text{ kJ/kgK}$$

$$\rho_w = 1000 \text{ kg/m}^3$$

$$m_w = \rho_w R_{t1} \times 10^{-6} \times 60 \quad (kg/min)$$

R_{t1} is the volume flow rate of water into the engine cooling jacket

Heat carried away by the exhaust gas

$$Q_e = m_{w1} c_{pw} (T_3 - T_1) \frac{T_4 - T_a}{T_4 - T_5} \quad (kJ/min)$$

Where,

m_{w1} is the mass flow rate of water into the calorimeter

T_3 is the outlet water temperature of calorimeter

T_1 is the inlet water temperature of calorimeter

T_4 is the exhaust gas temperature from the engine

T_5 is the exhaust gas temperature from the calorimeter

T_a is the ambient temperature

$$m_{w1} = \rho_w R_{t2} \times 10^{-6} \times 60 \quad (kg/min)$$

R_{t2} is the volume flow rate of water into the calorimeter.

Unaccounted heat or heat loss due to radiation

$$= Q_{in} - (Q_{bp} + Q_w + Q_e) \quad (kJ/min)$$

Experimentation:**Aim:**

To calculate heat balance sheet for single cylinder four stroke diesel engine.

Precautions:

1. Do not run the engine if supply voltage is less than 180V
2. Do not run the engine without the supply of water.
3. Supply water free from dust to prevent blockage in rotameters, engine head and calorimeter.
4. Note that the range for water supply provided is an approximate standard values, however the user may select the operating range to his convenience not less than 3 & 2 LPM for engine and calorimeter respectively.
5. Do not forget to give electrical earth and neutral connections correctly.
6. It is recommended to run the engine at 1500rpm otherwise the rotating parts and bearing of engine may run out.

Procedure:

- a. Give the necessary electrical connections to the panel.
- b. Check the lubricating oil level in the engine.
- c. Check the fuel level in the tank.
- d. Allow the water to flow to the engine and the calorimeter and adjust the flow rate to 6lpm & 3lpm respectively.
- e. Release the load if any on the dynamometer.
- f. Open the three-way cock so that fuel flows to the engine.
- g. Start the engine by cranking.
- h. Allow to attain the steady state.
- i. Load the engine by slowly tightening the yoke rod handle of the Rope brake drum.
- j. Note the following readings for particular condition,
 - i. Engine Speed
 - ii. Time taken for 10cc of diesel consumption
 - iii. Rotameter reading.
 - iv. Manometer readings, in cm of water &

- v. Temperatures at different locations.
- k. Repeat the experiment for different loads and note down the above readings.
- l. After the completion release the load and then switch off the engine.
- m. Allow the water to flow for few minutes and then turn it off.

Result:

4. HEAT BALANCE TEST ON FOUR STROKE FOUR CYLINDER DIESEL ENGINE

Introduction

A machine, which uses heat energy obtained from combustion of fuel and converts it into mechanical energy, is known as a Heat Engine. They are classified as External and Internal Combustion Engine. In an External Combustion Engine, combustion takes place outside the cylinder and the heat generated from the combustion of the fuel is transferred to the working fluid which is then expanded to develop the power. An Internal Combustion Engine is one where combustion of the fuel takes place inside the cylinder and converts heat energy into mechanical energy. IC engines may be classified based on the working cycle, thermodynamic cycle, speed, fuel, cooling, method of ignition, mounting of engine cylinder and application.

Diesel Engine is an internal combustion engine, which uses heavy oil or diesel oil as a fuel and operates on two or four stroke. In a 4-stroke Diesel engine, the working cycle takes place in two revolutions of the crankshaft or 4 strokes of the piston. In this engine, pure air is sucked to the engine and the fuel is injected with the combustion taking place at the end of the compression stroke. The power developed and the performance of the engine depends on the condition of operation. So it is necessary to test an engine for different conditions based on the requirement.

Specification

Engine	: Four stroke four cylinder water cooled diesel engine
Make	: Ambassador classic
Power	: 40 BHP @ 4000 rpm
Torque	: 8.6 kg-m @ 2000 rpm
Fuel	: Diesel
Bore	: 73 mm
Stroke length	: 88.9 mm

Observations:

Sl no.	Brake power BP (kW)	Speed N (rpm)	Flow rate of cooling water to engine jacket, R _{t1} (cc/sec)	Flow rate of cooling water to calorimeter, R _{t2} (cc/sec)

Sl no.	Thermocouple readings (°C)					Time for consuming 10cc of fuel, t (sec)
	T ₁	T ₂	T ₃	T ₄	T ₅	

Calculations:

Brake power

$$BP = \sqrt{3} VI \cos \phi = \text{_____} \text{ in kW}$$

Mass of fuel consumed

$$mfc = \frac{0.72 \times 3600 \times X}{1000 t} = \text{_____} \text{ kg/hr}$$

Specific fuel consumption

$$sfc = \frac{mfc}{BP} = \text{_____} \text{ kg/kW hr}$$

Heat input

$$Q_{in} = \frac{mfc \times CV}{60} = \text{_____} \text{ kJ/min}$$

Heat equivalent to brake power calculations

$$Q_{bp} = BP \times 60 = \text{_____} \text{ kJ/min}$$

Starting : Self start

Compression ratio : 23:1

Displacement volume : 1489 cc

Method of ignition : Compression ignition

Orifice diameter : 25 mm

Description of the apparatus:

The test rig is built for loading mentioned below:

1. The equipment consists of Ambassador Diesel Engine (self-started) of 40 BHP capacity and is Water cooled. The Engine is coupled to an AC generator for loading purposes. Coupling is done by an extension shaft in a separate bearing house.
2. Thermocouples are provided at appropriate positions and are read by a digital temperature indicator with channel selector to select the position.
3. Rota meters are used for direct measurement of water flow rate to the engine and calorimeter respectively.
4. Engine Speed at various conditions is determined by a Digital RPM.
5. A separate air box with orifice assembly is provided for regularizing and measuring the flow rate of air. The pressure difference at the orifice is measured by means of Manometer.
6. A volumetric flask with a fuel distributor is provided for measurement and directing the fuel to the engine respectively.
7. The engine test rig is directly coupled to an AC generator, 3 phase, 10kW, 50Hz, which in turn is loaded by resistance load bank. The load can be varied in steps of $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$ and full load by operating the rotary switches provided in the load bank panel

Thermocouple details

T₁ : Inlet water temperature of engine jacket and calorimeter

T₂ : Outlet water temperature of engine jacket

T₃ : Outlet water temperature from the calorimeter

T₄ : Exhaust gas temperature from engine

T₅ : Exhaust gas temperature from the calorimeter

Heat carried away by the engine cooling water

$$Q_w = m_w c_{pw} (T_2 - T_1) = \text{_____} \text{ kJ/min}$$

$$m_w = \rho_w R_{t1} \times 10^{-6} \times 60 = \text{_____} \text{ kg/min}$$

Heat carried away by the exhaust gas

$$Q_e = m_{w1} c_{pw} (T_3 - T_1) \frac{T_4 - T_a}{T_4 - T_5} = \text{_____} \text{ kJ/min}$$

$$c_{pw} = 4.187 \text{ kJ/kgK}$$

$$\rho_w = 1000 \text{ kg/m}^3$$

$$m_{w1} = \rho_w R_{t2} \times 10^{-6} \times 60 = \text{_____} \text{ kg/min}$$

Unaccounted heat or heat loss due to radiation

$$= Q_{in} - (Q_{bp} + Q_w + Q_e) = \text{_____} \text{ kJ/min}$$

Equations used:

Brake power

$$BP = \sqrt{3} VI \cos \phi \quad (\text{in kW})$$

Where,

V is the voltage across two phase

I is the current in each phase

ϕ is the power factor

Mass of fuel consumed

$$mfc = \frac{0.82 \times 3600 \times X}{1000 t} \quad (\text{kg/hr})$$

Where,

X is the burette reading in cc

0.82 is the density of diesel in gram per cc

t is the time taken in seconds

Specific fuel consumption

$$sfc = \frac{mfc}{BP} \quad (\text{kg/kW hr})$$

Heat input

$$Q_{in} = \frac{mfc \times CV}{60} \quad (\text{kJ/min})$$

CV is the calorific value of the fuel = 45200 kJ/kg

Heat equivalent to brake power calculations

$$Q_{bp} = BP \times 60 \quad (\text{kJ/min})$$

Heat carried away by the engine cooling water

$$Q_w = m_w c_{pw} (T_2 - T_1) \quad (\text{kJ/min})$$

Where,

c_{pw} is the specific heat of water

ρ_w is the density of water

m_w is the mass flow rate of water into the engine cooling jacket

T_2 is the outlet water temperature of engine water jacket

T_1 is the inlet water temperature of engine water jacket

Heat balance sheet

Sl. No.	Particulars	Heat Content kW	%
1	Heat Input -- A		100
2	Heat to BP -- B		B/A =
3	Heat to Cooling Water -- C		C/A =
4	Heat to Exhaust Gases -- D		D/A =
5	Heat Unaccounted -- E		E/A =

$$c_{pw} = 4.187 \text{ kJ/kgK}$$

$$\rho_w = 1000 \text{ kg/m}^3$$

$$m_w = \rho_w R_{t1} \times 10^{-6} \times 60 \quad (\text{kg/min})$$

R_{t1} is the volume flow rate of water into the engine cooling jacket

Heat carried away by the exhaust gas

$$Q_e = m_{w1} c_{pw} (T_3 - T_1) \frac{T_4 - T_a}{T_4 - T_5} \quad (\text{kJ/min})$$

Where,

m_{w1} is the mass flow rate of water into the calorimeter

T_3 is the outlet water temperature of calorimeter

T_1 is the inlet water temperature of calorimeter

T_4 is the exhaust gas temperature from the engine

T_5 is the exhaust gas temperature from the calorimeter

T_a is the ambient temperature

$$m_{w1} = \rho_w R_{t2} \times 10^{-6} \times 60 \quad (\text{kg/min})$$

R_{t2} is the volume flow rate of water into the calorimeter.

Unaccounted heat or heat loss due to radiation

$$= Q_{in} - (Q_{bp} + Q_w + Q_e) \quad (\text{kJ/min})$$

Experimentation:

Aim:

To calculate heat balance sheet for four cylinder four stroke diesel engine.

Precautions:

1. Do not run the engine if supply voltage is less than 180V
2. Do not run the engine without the supply of water.
3. Supply water free from dust to prevent blockage in rotameters, engine head and calorimeter.

4. Note that the range for water supply provided is an approximate standard values, however the user may select the operating range to his convenience not less than 3 & 2 LPM for engine and calorimeter respectively.
5. Do not forget to give electrical earth and neutral connections correctly.
6. It is recommended to run the engine at 1500rpm otherwise the rotating parts and bearing of engine may run out.

Procedure:

- a. Give the necessary electrical connections to the panel.
- b. Check the lubricating oil level in the engine.
- c. Check the fuel level in the tank.
- d. Allow the water to flow to the engine and the calorimeter and adjust the flow rate to 6lpm & 3lpm respectively.
- e. Release the load if any on the dynamometer.
- f. Open the three-way cock so that fuel flows to the engine.
- g. Start the engine by cranking.
- h. Allow to attain the steady state.
- i. Load the engine by operating the rotary switches on the resistance of the load bank, in steps of $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, and full load.
- j. Note the following readings for particular condition,
 - i. Engine Speed
 - ii. Time taken for_____cc of diesel consumption
 - iii. Rotameter reading.
 - iv. Manometer readings, in cm of water &
 - v. Temperatures at different locations.
- k. Repeat the experiment for different loads and note down the above readings.
- l. After the completion release the load and then switch off the engine.
- m. Allow the water to flow for few minutes and then turn it off.

Result:

5. MORSE TEST ON 3 - CYLINDER 4 -STROKEPETROL ENGINE

Introduction

A machine, which uses heat energy obtained from combustion of fuel and converts it into mechanical energy, is known as a Heat Engine. They are classified as External and Internal Combustion Engine. In an External Combustion Engine, combustion takes place outside the cylinder and the heat generated from the combustion of the fuel is transferred to the working fluid which is then expanded to develop the power. An Internal Combustion Engine is one where combustion of the fuel takes place inside the cylinder and converts heat energy into mechanical energy. IC engines may be classified based on the working cycle, thermodynamic cycle, speed, fuel, cooling, method of ignition, mounting of engine cylinder and application.

Description of the apparatus:

a) Hydraulic Dynamometer Loading

- 1) The equipment consists of a Maruti 800 Engine (Self started) of capacity 800cc and power 12HP.
- 2) The Engine is coupled to a Hydraulic Dynamometer for loading purposes. The coupling is done by a universal coupling in a bearing house.
- 3) Thermocouples are provided at appropriate positions and are read by a digital temperature indicator with channel selector.
- 4) Rotameters are used for direct measurement of water flow rate to the engine and calorimeter respectively.
- 5) The Engine Speed and the load applied at various conditions is determined by a Digital RPM Indicator and Load cell arrangement.
- 6) A separate air box with orifice assembly is provided for regularizing and measuring the flow rate of air. The pressure difference is measured by means of a Manometer.
- 7) A burette with a fuel distributor is provided for measurement and directing the fuel to the engine respectively.

Observations:

Running cylinder nos.	Cut-off cylinder number	Load W (kg)	Engine speed N (rpm)	Brake power of running cylinder (kW)	Indicated power of the cut-off cylinder (kW)
1,2 and 3					-
2 and 3	1				
1 and 3	2				
1 and 2	3				

Calculations:

Brake power

$$BP = \frac{W \cdot N}{C} = \text{_____} kW$$

Indicated power of each cylinder

$$IP_1 = BP_{total} - BP_{2\&3} = \text{_____} kW$$

$$IP_2 = BP_{total} - BP_{1\&3} = \text{_____} kW$$

$$IP_3 = BP_{total} - BP_{1\&2} = \text{_____} kW$$

$$IP_{total} = IP_1 + IP_2 + IP_3 = \text{_____} kW$$

Frictional power

$$FP = IP_{total} - BP_{total} = \text{_____} kW$$

Brake thermal efficiency

$$\eta_{bth} = \frac{BP \times 3600 \times 100}{mfc \times CV} = \text{_____}$$

$$CV = 43500 \text{ kJ/kg}$$

Mechanical efficiency

$$\eta_{mech} = \frac{BP}{IP} = \text{_____}$$

Engine specification

Engine	: Maruti 800
BHP	: 12HP
Speed	: 2000rpm
Fuel	: Petrol
No. of cylinders	3
Bore	: 68.5mm
Stroke length	: 72mm
Starting	: Self start
Working cycle	: four stroke
Method of cooling	: water cooled
Method of ignition	: spark ignition

Instrumentation

Digital RPM indicator to measure the speed of the engine.

Digital temperature indicator to measure temperatures at various points.

Differential manometer to measure quantity of air sucked into cylinder.

Burette with stop cock to measure the rate of fuel consumed.

Spring balance to measure load applied to the engine.

b) Fuel measurement

The fuel is supplied to the engine from the main fuel tank through a graduated measuring fuel gauge (Burette) with 2-way cock. To measure the fuel consumption of the engine, fill the burette by opening the cock. By starting a stop clock, measure the time taken to consume 10cc of fuel by the engine.

c) Air intake measurement

The suction side of the engine is connected to an Air tank. The atmospheric air is drawn into the engine cylinder through the air tank. The manometer is provided to measure the pressure drop across an orifice provided in the intake pipe of an air tank. This pressure drop is used to calculate the volume of air drawn into the cylinder. (Orifice diameter is 25mm)

d) Lubrication

The engine is lubricated by mechanical lubrication.

Lubricating oil recommended – SAE – 40 or equivalent.

e) Temperature measurement

A digital temperature indicator with selector switch is provided on the panel to read the temperature in degree centigrade, directly sensed by respective thermocouples located at different places on the test rig.

Note: use the selector knob gently.

Thermocouple details

T₁ : Inlet water temperature of engine jacket and calorimeter

T₂ : Outlet water temperature of engine jacket

T₃ : Outlet water temperature from the calorimeter

T₄ : Exhaust gas temperature from calorimeter

f) Water flow measurement

A flowmeter is provided at the inlet jacket to measure the quantity of water allowed into the engine jacket. A rotameter is provided at inlet of calorimeter. Valves are provided to regulate the flow rate of water. Readings can be taken from flow meter in litres and in rotameter in cc/sec.

g) Loading system

The engine is loaded by varying the quantity of water allowed into the dynamometer at constant pressure head. We can vary the load by operating the gate valve provided on the inlet line of the dynamometer. The dynamometer outlet at the bottom can be adjusted depending upon the load conditions.

Experimentation:

Aim:

The experiment is conducted to find the indicated power of each cylinder using Morse test.

Equations used:

Brake power

$$BP = \frac{W \cdot N}{C} \quad (\text{in kW})$$

Where,

W is the spring balance reading in kg

N is the speed of engine in rpm

C is a constant, $C = 3240$

Indicated power of each cylinder

$$IP_1 = BP_{total} - BP_{2\&3} \quad (in \text{ kW})$$

$$IP_2 = BP_{total} - BP_{1\&3} \quad (in \text{ kW})$$

$$IP_3 = BP_{total} - BP_{1\&2} \quad (in \text{ kW})$$

$$IP_{total} = IP_1 + IP_2 + IP_3 \quad (in \text{ kW})$$

Frictional power

$$FP = IP_{total} - BP_{total} \quad (in \text{ kW})$$

Brake thermal efficiency

$$\eta_{bth} = \frac{BP \times 3600 \times 100}{mfc \times CV}$$

Where,

CV is the calorific value of petrol = 43500kJ/kg

BP is the brake power in kW

Mechanical efficiency

$$\eta_{mech} = \frac{BP}{IP}$$

Where,

IP is the indicated power in kW

Procedure:

1. Run the engine at 2000rpm and load 12kg. Morse test can be conducted by disconnecting the power of the individual cylinder one by one with the use of knife switches provided on the panel.
2. Cut off one cylinder, then the engine speed will drop. Bring back the speed of the engine to 2000rpm by reducing the load and measure the power developed.
3. Repeat the above steps by cutting of the cylinders 2 and 3.
4. Calculate the break power with 3 cylinders at 2000rpm and then calculate the brake power with one cylinder cut off. The difference will give indicated power of one cylinder, i.e. cut off cylinder. Similarly calculate the indicated power of

the remaining two cylinders. Then the total indicated power of the engine can be calculated by adding the indicated power of the individual cylinder.

5. To find out friction power of the engine, deduct total brake power from the total indicated power.
6. After the completion release the load (while doing so release the accelerator) and then switch of the engine and the panel.
7. *Allow the water to flow for few minutes and then turn it off.

*Note:

Allow water only to the calorimeter and not to the engine.

8. Remove earthing connection of the battery and the starter key.

Precautions:

1. Do not run the engine if supply voltage is less than 180V
2. Do not run the engine without the supply of water.
3. Supply water free from dust to prevent blockage in rotameters, engine head and calorimeter.
4. Note that the range for water supply provided is an approximate standard values, however the user may select the operating range to his convenience not less than 3 & 2 LPM for engine and calorimeter respectively.
5. Always set the accelerator knob to the minimum condition and start the engine.
6. Do not forget to give electrical earth and neutral connections correctly.
7. It is recommended to run the engine below 3000rpm otherwise the rotating parts and bearing of dynamometer may run out.

Result: